

5.5 Equations and Solutions in the Real World

Lesson Introduction

 Benchmarks	MA.912.AR.2.1 Given a real-world context, write and solve one-variable, multi-step linear equations.		 Learning Targets		- I can write and solve multi-step equations in a real-world context. - I can interpret solutions as viable or non-viable in a given context.
 Benchmark Coherence	Building On MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.		Working Toward N/A		
 Essential Questions	- How do I write and solve multi-step equations in a real-world context? - How can I determine if a solution is viable or non-viable in a given context?				
 Lesson Narrative	In this lesson, students progress from writing and solving two-step equations representing real-world contexts to writing and solving multi-step equations representing real-world contexts. Students will learn how to write expressions that represent different components of a word problem and bring them together to write a multi-step equation that represents the real-world context, then use that equation to solve for the variable.				
 Component Summary	5.5.1 Warm-Up (~5 min)	Which One Doesn't Belong activity (<i>SE p. 249</i>)	5.5.3 Guided Practice (10-15 min)	Practice writing and solving equations representing real-world contexts (<i>SE p. 251</i>)	
	5.5.2 Guided Instruction (10-15 min)	Introduction to writing and solving equations representing real-world contexts (<i>SE p. 249</i>)	5.5.4 Your Turn! (10 min)	Independent practice writing and solving equations representing real-world contexts (<i>SE p. 252</i>)	
	5.5.5 Wrap-Up (~5 min)	Reflection on learning targets (<i>SE p. 253</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		(Optional) Different colored highlighters		N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	Students might attack problems without formulating a plan for solving them, most often writing coefficients as constants and constants as coefficients. For example, in question 2 in 5.5.3 Guided Practice, some students may identify per person or per bag as a constant.	Utilize 5.5.2 Guided Instruction to model the reading strategy for students to leverage throughout the lesson. Text-heavy lessons, like Lesson 5, require decoding strategies for 100% of students to be engaged, 100% of the time.
	Literacy and Language Routines Three Reads This lesson is comprised completely of real-world problems. For this lesson, consider leveraging the Three Reads routine. This routine supports reading comprehension of problems and meta-awareness of mathematical language.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.7.1: Real-World Contexts This lesson consists solely of real-world problems. During this lesson, students will connect mathematical concepts to everyday experiences. These problems will provide opportunities to create models, both concrete and abstract, and perform investigations.
 Differentiate Instruction	Consider a variety of strategies for decoding and interpreting real-world contexts. Leverage anchor charts or similar visual prompts that students can use as scaffolding throughout the lesson. - Three Reads: Give each read a specific purpose to “chunk” or scaffold the decoding process - Mark-Up Text: Visually identify and highlight important information for problem-solving - Partner Reading: Opportunity for students to collaborate, ask questions, and balance strengths	
Lesson Notes:		

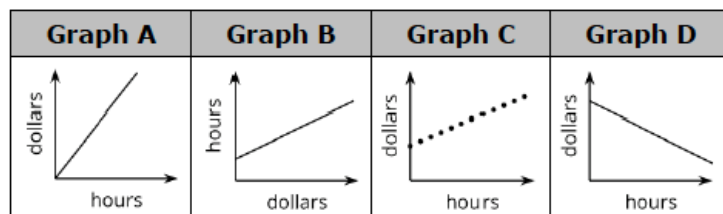
5.5.1 Warm-Up: Which One Doesn't Belong?

Component Overview: Students will review four graphs to determine which graph doesn't belong.



5.5.1 Warm-Up: Which One Doesn't Belong?

Four graphs are shown.



Which One Doesn't Belong

This instructional routine purposefully has no right or wrong answer. This routine facilitates a classroom in which process matters more than product, creativity is highlighted, and unique answers are encouraged.

- Complete the statement to explain which graph does not belong.

Answers will vary.

- ☐ Graph A
- ☐ Graph B
- ☐ Graph C
- ☐ Graph D

does not belong because the graph has a negative slope.

- Find a classmate who selected a different graph that does not belong. Write their explanation in the table. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary of their explanation stating that your summary is correct.

My Summary of _____'s Description	Initials
<i>Answers will vary.</i>	

5.5.2 Guided Instruction: Writing and Solving Equations Representing Real-World Contexts

Component Overview: Students will be introduced to writing and solving equations representing real-world contexts by using two real-world problems.



5.5.2 Guided Instruction: Writing and Solving Equations Representing Real-World Contexts

1. The school choir is taking a day trip to Volcano Bay. They got a special ticket rate of \$65 per person, but the cost of the bus is \$639.90. If they fill every seat on the bus, then each person will pay a total of \$76.85 for bus transportation and an admission ticket.
 - a. Complete the table to write two different algebraic expressions for the cost of this trip, where s represents the number of people on the trip.

Expression A	Expression B
<i>ticket rate per person plus the total bus cost</i>	<i>total amount paid per person, if all seats are full</i>
$65s + 639.90$	$76.85s$

If all the seats on the bus are full, then Expression A and Expression B will be equivalent to each other. Therefore, they can be set equal to each other to create an equation.

- b. Write and solve this equation.

$$65s + 639.90 = 76.85s$$

$$639.90 = 11.85s$$

$$54 = s$$
 - c. Interpret the solution in context.

54 is the number of students needed to fill the bus.
 - d. Discuss with your partner why 52.7 would not be a viable solution for this context. Summarize your discussion.

It isn't possible to have a partial person.



Mark-Up Text or Three Reads

Model one of your decoding strategies here for students to utilize throughout the lesson. Consider using visual aids and multiple colors to highlight key information from written text in the problem to mathematical translation into an expression. Use different colors to highlight key words and phrases with their corresponding translations, such as expression A and expression B in question 1a.



What does that mean about the type of solutions that would be considered viable?

5.5.2 Guided Instruction: Writing and Solving Equations Representing Real-World Contexts (continued)

2. Max and his friends are planning a chartered fishing trip off the coast of Fort Lauderdale. Fishing Adventures charges a base fee of \$299.22, plus \$21.90 per person for a 4-hour trip. However, Max found a 10% off coupon for Ocean Excursions. Normally Ocean Excursions offers 6-hour trips for \$50.58 per person plus a flat fee of \$70 for equipment. With the 10% off coupon, the two options would cost Max and his friends the same amount of money.

- a. Define a variable to represent the unknown.
p, the number of people on the fishing trip.

- b. Write and solve an equation representing this situation.

$$21.90p + 299.22 = 0.90(50.58p + 70)$$

$$21.90p + 299.22 = 45.522p + 63$$

$$299.22 = 23.622p + 63$$

$$236.22 = 23.622p$$

$$10 = p$$

- c. Interpret the solution.

Answers will vary. There are 10 people going on the fishing trip. Max is going on the trip with 9 of his friends.



Three Reads

When reviewing these problems, encourage students to engage in the Three Reads instructional routine. This routine supports reading comprehension of problems and meta-awareness of mathematical language.

- The first read is to understand the context. What is happening here?
- The second read is to understand the mathematics. What information is here?
- The third read is to generate a plan for solving the problem. How can the information from the second read help answer the first read?



- Is this value a viable solution for this context?
- For 10 people, which fishing trip is the better deal?

5.5.3 Guided Practice: Solving Real-World Problems by Writing and Solving Equations

Component Overview: Students will practice solving two real-world problems by writing and solving equations.



5.5.3 Guided Practice: Solving Real-World Problems by Writing and Solving Equations

1. Marisol was surprised when she discovered last week that she had worked the same number of hours and made the same amount of money by working two different jobs. As a server at The Main Street Grill, she earned \$5.85 per hour plus \$74.40 in tips. Working at Fresh Fashions she earned \$8.65 per hour plus \$22.60 in sales bonuses.
 - a. Define a variable to represent the unknown.
 h = the number of hours worked per job
 - b. Write an expression that represents the amount Marisol earned working at The Main Street Grill.
 $5.85h + 74.40$
 - c. Write an expression that represents the amount Marisol earned working at Fresh Fashions.
 $8.65h + 22.60$
 - d. Write the equation that represents last week, when she earned the same amount of money working the same number of hours at both jobs.
 $5.85h + 74.40 = 8.65h + 22.60$
 $74.40 = 2.8h + 22.60$
 $51.80 = 2.8h$
 $18.50 = h$
 - e. Solve the equation.
 $5.85h + 74.40 = 8.65h + 22.60$
 $74.40 = 2.8h + 22.60$
 $51.80 = 2.8h$
 $18.50 = h$
 - f. Interpret the solution.
She worked 18.5 hours at each job last week. (She worked a total of 37 hours.)



Consider leveraging a Think-Pair-Share instructional routine for question 1. Allow students some individual think time to process the problem and work through questions 1a-1h. Then have students work with a partner to share their answers and come to consensus on the correct answer for each of the questions.



How can you confirm that the answer to question 1e is the correct solution to the equation?

5.5.3 Guided Practice: Solving Real-World Problems by Writing and Solving Equations (continued)

- g. Explain whether the mathematical solution to the equation is viable in this context.

It is viable because she can work a partial hour.

- h. Complete the statement. Use the space provided below to show any work.

Last week, Marisol worked a total of 37 hours and earned a total of \$365.25.

2. The Zamora family of four is flying from Orlando to New York's JFK airport. They found tickets with Sky Airlines for \$74 per person round trip, but there is a \$25 charge per bag. Mrs. Zamora found that for the total number of bags they plan to bring, they can save money by upgrading their tickets. With the upgrade, they would pay \$103 per person for the tickets, \$5 per bag, and they would still save \$4 total.

- a. Define a variable to represent the unknown.

b = the number of bags the family is bringing

- b. Write and solve an equation that represents this situation.

$$25b + 74(4) = 5b + 103(4) + 4$$

$$25b + 296 = 5b + 412 + 4$$

$$20b + 296 = 416$$

$$20b = 120$$

$$b = 6$$

- c. Interpret the solution.

The family is planning to bring 6 bags on the trip.



- What information from this word problem is important?
- What is this word problem asking you to solve?



Is this solution viable for the context? Why or why not?

5.5.4 Your Turn!

Component Overview: Students will solve one real-world problem by writing and solving equations independently.



5.5.4 Your Turn!

1. The student council at Cypress Creek High School had a jog-a-thon fundraiser with a designated pledge amount per mile. Each class can raise additional money through donations of any amount. The freshman and sophomore classes raised the same amount of money. The freshman class ran a total of 562 miles and collected \$131.25 in donations. The sophomore class collected \$185 in donations and ran a total of 519 miles.

- a. Define a variable to represent the unknown.

x = the amount of money pledged per mile

- b. Write and solve an equation that represents this situation.

$$562x + 131.25 = 519x + 185$$

$$43x + 131.25 = 185$$

$$43x = 53.75$$

$$x = 1.25$$

- c. Interpret the solution.

The pledge amount is \$1.25 per mile.

- d. How much money did each class raise?

\$833.75



Consider using this time to work with individual students or small groups based on those who were struggling during the 5.5.3 Guided Practice.



Is this solution viable for the context? Why or why not?

5.5.5 Wrap-Up: Reflection

Component Overview: Students complete a self-reflection on their level of understanding of the concepts associated with solving equations.



5.5.5 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each.

Answers will vary.



I need help.



I got it.



I could help someone.

Identifying and defining the variable for a real-world context	  
Writing a multi-step equation to represent a context	  
Solving multi-step one-variable linear equations	  
Interpreting the meaning of the solution in a given context	  



Prompt students to explain their choices using any creative method. They can write a response, use the icons to mark-up the lesson text, or other creative methods like recording or helping a classmate.